

Exploring Processes Using Sysinternals

Sysinternals is a freeware tool that can help to manage, diagnose, troubleshoot and monitor a Windows environment. It is a part of the Microsoft TechNet website which offers diagnostic tools, technical resources and utilities. This article briefly explains how Sysinternals can be used to give information about running processes.



You can now use the Sysinternals tools to find out which program has a particular file or directory open. The tools available in the Sysinternals suite show details about running process handles and DLLs. The entire suite can be downloaded from <https://download.sysinternals.com/files/SysinternalsSuite.zip>

Process monitor

Process monitor is a free utility available to monitor real-time file systems, registry, process/threads and the network. It can be downloaded from <https://technet.microsoft.com/en-us/library/bb896645.aspx>.

Trace summary tools

Process monitor includes a number of dialogues that allow you to perform simple data mining on the events collected in a trace.

Process tree displays all of the processes referenced in a hierarchy in the loaded trace, which shows parent-child relationships. Go to **Tools > Process Tree** or **Ctrl+T**.

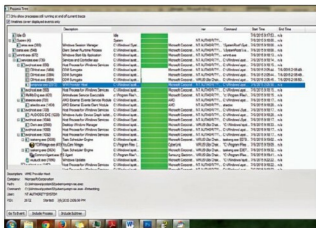


Figure 1: Process tree for various processes

File system

Process monitor displays all the activities of a file system, including local and remote storage. It also detects and monitors new file system devices.

File summary

File summary dialogue lists each unique file system path present in the filtered trace, the amount of time spent performing I/O to the file, the total number of events that referenced the path, and the count of individual operation types.

File summary can be accessed by going to **Tools > File summary**.

The file summary can be achieved by means of the folder and extension.

Activity summary

Activity summary lists all the processes seen in the trace, file events, I/O, registry events, network events, including their process ID, image name and command line.

Activity summary can be accessed by going to **Tools > Process Activity Summary**.

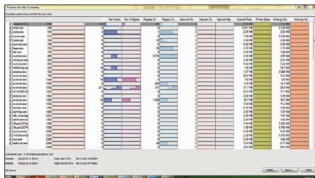


Figure 4: Activity summary for various processes and their operations